EXPERIMENT

Redox titration of hypochlorite in bleach

Goal

The goal of this laboratory is to perform an **oxidation-reduction titration** to determine the concentration of sodium hypochlorite (NaOCl) in commercial bleach samples.

Materials

- ☐ A buret, a stad and a clamp
- ☐ 100mL, 50mL beakers
- ☐ 125mL Erlenmeyer
- ☐ 0.025M $Na_2S_2O_3$ solution
- ☐ solid KI
- ☐ 10% diluted bleach
- ☐ 10mL transfer pipet
- ☐ 2M HCl solution
- ☐ 0.2% starch solution

Background

The properties of chlorine for bleaching fabrics were discovered in the 18th century by the French scientist Claude Berthollet. He was able to make an aqueous solution of sodium hypochlorite which was named after the quarter in Paris where it was produced, Javel. *Eau de Javel* (Javel water) is still used in French to name bleach.

At the beginning of the 18th century, the pharmacist Antoine Germain Labarraque discovered the disinfecting ability of another hypochlorite solution, this time calcium hypochlorite. The so-called "Eau de Labarraque" was employed in the disinfection of animal guts to make musical instrument strings. Hypochlorite was the first antiseptic product, helping to treat gangrene and putrescent wounds in people in the 1820s, and ending with the propagation of "cadaveric particles" in morgues.

Nowadays bleach is a common product at home, used to whiten clothes, remove stains, and as a disinfectant. Because of its strong bactericidal properties, hypochlorite is also used for disinfecting and to prevent the proliferation of algae in swimming pools. Bleach can be consumed. The CDC recommends adding eight drops of bleach per gallon of clear water. For cloudy water, the bleach quantity should be doubled to 16 drops per gallon. Bleach can be found in powder form as well, in the form of Bleaching powder. This is indeed, a solid combination of chlorine and slaked lime, introduced in 1799 by Scottish chemist Charles Tennant.

Chlorine

Chlorine forms four different polyatomic anions with oxygen:

ClO^- (hypochlorite) ClO_2^- (chlorite) ClO_3^- (chlorate) ClO_4^- (perchlorate)

These are called **oxoanions** and illustrate 4 possible oxidation states of chlorine atoms; +1, +3, +5, and +7 respectively. When the oxidation state of the chlorine decreases, for instance going from ClO_4^- to ClO_3^- ($+7 \rightarrow +5$), it reduces. For reduction to happen another species must be oxidized. Chlorine is then acting as the **oxidizing agent**. Perchlorate it is the strongest oxidizing agent of the 4 oxoanions, but even ClO^- can still be reduced to Cl- (Cl oxidation state -1).

The first oxidation

To determine the concentration of ClO^-, two consecutive oxidation-reduction reactions will be performed. In the first reaction, all the ClO^- present in the sample will be oxidized into Cl^-. The **oxidizing agent** will be I^-, which will be in turn reduced into I_2. The reaction, carried out in an acidic medium, will read as:

$$H^+_{(aq)} + ClO^-_{(aq)} + I^-_{(aq)} \longrightarrow I_{2\,(s)} + Cl^-_{(aq)} + H_2O_{(l)} \qquad ⚠ \text{ (unbalanced)}$$

The reaction above is not balanced. You can determine the number of electrons being transferred using the oxidation states to help you balance the reaction. An excess amount of I^- will be used to ensure that all the ClO^- is oxidized. The iodine (I_2) formed in this reaction will be therefore limited by the amount of hypochlorite present in the sample. Keep in mind that the amount of hypochlorite is what we want to determine, and it can be stoichiometrically calculated from the amount of iodine produced. In the following step, the amount of I_2 will be determined using an oxidation-reduction titration. The solution might turn slightly brownish due to the presence of a Lewis complex between iodine and water.

The oxidation-reduction titration

The titration to be performed accounts for the following redox reaction:

$$I_{2\,(s)} + S_2O_3{}^{2-}{}_{(aq)} \longrightarrow I^-_{(aq)} + S_4O_6{}^{2-}{}_{(aq)} \qquad ⚠ \text{ (unbalanced)}$$

Again, this reaction is not balanced. Iodine (I_2) is reduced back to iodide I^-. Thiosulfate $S_2O_3{}^{2-}$is the reducing agent, being itself oxidized into tetrathionate ions ($S_4O_6{}^{2-}$). The titration endpoint will be determined by adding starch as an indicator at the beginning of the reaction. Remember from previous experiments that starch and I_2 react turning the solution into a dark blue color, therefore, the titration should be stopped when all the iodine has been reduced and the solution becomes transparent.

Redox reactions

Example

Calculate the redox number of the elements underlined in the following molecules: (a) $K_2\underline{C}O_3$ and (b) $H_2\underline{C}O$.

Answer: Let us set up the redox equation for the first compound, knowing that the redox of oxygen is −2 and potassium +1. The unknown variable x represents the redox number of the underlined element. We have:

$$2 \cdot (+1) + x + 3 \cdot (-2) = 0$$

Mind we have two potassium and three oxygens hence we need to time the redox of K by two and the redox of O by three. If we solve for x we obtain a redox number for carbon of IV. The redox equation for the second example is:

$$2 \cdot (+1) + x + (-2) = 0$$

Mind that according to the redox rules, the redox number of oxygen is +1. Solving for x we have a redox number of zero.

Example

Balance the following redox in acidic medium:

$$Cr_2O_7{}^{2-} + SO_3{}^{2-} \longrightarrow Cr^{3+} + SO_4{}^{2-}$$

Answer:
We locate the same element in both sides of the reaction with different redox number. We found Chromium in the form of dichromate ($Cr_2O_7{}^{2-}$) with redox number VI and Chromium in the product side with redox III. Therefore Chromium is being reduced, as its redox number decreases. Differently, we found sulphur in the reactants side in the form of sulfite ($SO_3{}^{2-}$) with redox number of IV and in the product side with redox number of VI. Therefore Sulphur is being oxidized. We will first set up the oxidation half-reaction knowing that we have different amounts of Cr in both side and that we will have to add water molecules to balance O and protons to balance H. As we have seven oxygens in $Cr_2O_7{}^{2-}$ we will

have to add seven water molecules. Also, as we add seven water molecules we will have to add fourteen protons. We will need six electrons to compensate the charge:

$$Cr_2O_7{}^{2-} \quad + \quad 14\,H^+ \quad + \quad 6\,e^- \quad \longrightarrow \quad 2\,Cr^{3+} \quad + \quad 7\,H_2O \text{ (reduction)}$$

For the reduction half-reaction, we have a difference of one oxygen atoms and hence we will need one water molecule and two protons; we will need two electrons to compensate the charge:

$$SO_3{}^{2-} \quad + \quad H_2O \quad \longrightarrow \quad SO_4{}^{2-} \quad + \quad 2\,H^+ \quad + \quad 2\,e^- \text{ (oxidation)}$$

In order to add both half-reactions as the reduction involves two electrons and the oxidation six, we will have to multiply the reduction half-reaction by three:

$$\begin{array}{ll}
1\cdot\left(\,Cr_2O_7{}^{2-} + 14\,H^+ + 6\,e^- \longrightarrow 2\,Cr^{3+} + 7\,H_2O\,\right) & \text{(reduction)} \\
3\cdot\left(SO_3{}^{2-} + H_2O \longrightarrow SO_4{}^{2-} + 2\,H^+ + 2\,e^-\right) & \text{(oxidation)} \\
\hline
Cr_2O_7{}^{2-} + 14\,H^+ + 3\,SO_3{}^{2-} + 3\,H_2O + \cancel{6\,e^-} \longrightarrow & \text{(redox)} \\
\qquad 2\,Cr^{3+} + 7\,H_2O + 3\,SO_4{}^{2-} + 6\,H^+ + \cancel{6\,e^-} &
\end{array} +$$

After we cancel the electrons and eliminate protons and water molecules, the balanced reaction is:

$$8\,H^+\,(aq) + \overset{VI}{Cr}_2O_7^{2-}\,(aq) + 3\,\overset{IV}{S}O_3^{2-}\,(aq) \longrightarrow 3\,\overset{VI}{S}O_4^{2-}\,(aq) + 2\,\overset{III}{Cr}{}^{3+}\,(aq) + 7\,H_2O_{(l)}$$

$$\text{Oxidation:} -2e^- \qquad \text{Reduction:} +3e^-$$

The experiment

The experimental part consists of a trial titration and an exact titration. The trial titration will be used to estimate the volume added at the endpoint of the titration. It is conducted quickly, adding portions of 1 mL until the endpoint is exceeded. The final result is not used for calculations. In the exact titration, an initial amount is added rapidly. This amount is calculated by subtracting 5 mL from the volume used in the trial titration. Like this it is possible to stop before the endpoint of the titration and to continue carefully, adding the titrant on a drop-by-drop basis until the precise endpoint. If time allows, the exact titration should be repeated and the mean volume used in calculations.

Procedure

Part A. Trial titration.

☐ *Step 1:* – Obtain a solution of bleach (the instructor might give you an unknown bleach sample, if so write down the Unknown # in the results section).

☐ *Step 2:* – Obtain a buret, a stand, and a clamp. Rinse the buret with distilled water and make sure the valve works properly.

☐ *Step 3:* – In a 100 mL beaker, obtain 70 mL of 0.025 M solution of $Na_2S_2O_3$. Use a small portion of the solution to rinse the buret. Discard the rinsing liquid and fill up the buret above the zero mark. Bring the volume in the buret to zero or below, discarding the excess liquid through the tip of the buret.

☐ *Step 4:* – In a 125 mL Erlenmeyer, obtain about one inch of a spatula of KI.

⚠CAUTION!

⚠ Bleach is a base.

⚠ Bases and acids can cause burns and clothes color loss.

⚠ Always wear eye protection and handle carefully.

☐ *Step 5:* – In a 50 mL beaker, obtain about 30 mL of a 10% bleach solution.

☐ *Step 6:* – Obtain a 10 mL transfer pipet. Pipet 10.0 mL of the bleach into the KI containing Erlenmeyer. Add 20 mL of distilled water from a graduated column and 20 drops of HCl 2 M. Homogenize the mixture by stirring the flask.

☐ *Step 7:* – Record the buret initial reading to the closest hundredth of mL.

☐ *Step 8:* – Place the Erlenmeyer under the buret and start the titration by adding 1 mL portions until the solution turns from brown to yellow.

☐ *Step 9:* – Add 40 drops of a 0.2% starch solution to the yellow solution in the Erlenmeyer.

☐ *Step 10:* – Proceed with the titration adding the $Na_2S_2O_3$ in 1 mL portions until it turns from dark color to colorless.

☐ *Step 11:* – Record the buret final reading and calculate the volume added.

☐ *Step 12:* – Discard the solution in the Erlenmeyer in the corresponding waste container.

Part B. Exact titration.

☐ *Step 1:* – Refill the buret with $Na_2S_2O_3$ 0.025 M. At this point you know how much volume you will need from the trial titration. You don not need to fill the buret up to zero but add a few milliliters more than the expected volume.

☐ *Step 2:* – In a 125 mL Erlenmeyer, obtain about one inch of a spatula of KI.

☐ *Step 3:* – Pipet 10.0 mL of the diluted bleach into the KI containing Erlenmeyer. Add 20 mL of distilled water from a graduated column and 20 drops of HCl 2 M. Homogenize the mixture by stirring the flask.

☐ *Step 4:* – Record the buret initial reading to the closest hundredth of mL.

☐ *Step 5:* – Place the Erlenmeyer under the buret and add 3 mL less than the volume used in the trial titration.

☐ *Step 6:* – Add 40 drops of a 0.2% starch solution to the solution in the Erlenmeyer.

☐ *Step 7:* – Continue the titration adding the titrant drop by drop until the solution turns transparent.

☐ *Step 8:* – Record the buret final reading in your notes.

☐ *Step 9:* – Discard the solution in the Erlenmeyer. If the time allows, do two exact titration replicates.

☐ *Step 10:* – When finished empty the buret and rinse it with plenty of distilled water. The buret should be stored empty, open, and upside down.

Calculations

1 This is the initial reading of the buret (often is zero if the buret is all filled up).

2 This is the final reading of the buret after the endpoint has been reached.

3 This is the volume of titrant used, $v_{S_2O_3{}^{-2}}$:

$$v_{S_2O_3{}^{-2}} = 2 - 1$$

4 This is the moles of thiosulfate used in the titration:

$$n_{S_2O_3{}^{-2}} = 3 \times 0.025 \times 10^{-3}$$

5 These are the moles of bleach in the (diluted) sample:

$$n_{OCl^{-1}} = \frac{n_{S_2O_3{}^{-2}}}{2} = \frac{4}{2}$$

6 This is the molarity of bleach in the (diluted) sample:

$$c^{diluted}_{OCl^{-1}} = \frac{n_{OCl^{-}}}{1 \times 10^{-2}} = \frac{5}{1 \times 10^{-2}}$$

7 This is the molarity of bleach in the (original, concentrated) sample:

$$c^{concentrated}_{OCl^{-1}} = c^{diluted}_{OCl^{-1}} \times 10 = 6 \times 10$$

STUDENT INFO

Name: ____________________ Date: __________

Pre-lab Questions

Redox titration of hypochlorite in bleach

1. Balance the following redox reaction, indicating the oxidizing and reducing agents:

$$\_\_H^+_{(aq)} + \_\_ClO^-_{(aq)} + \_\_I^-_{(aq)} \longrightarrow \_\_I_{2\,(s)} + \_\_Cl^-_{(aq)} + \_\_H_2O_{(l)}$$

2. Balance the following redox reaction, indicating the oxidizing and reducing agents:

$$\_\_I_{2\,(s)} + \_\_S_2O_3{}^{2-}{}_{(aq)} \longrightarrow \_\_I^-_{(aq)} + \_\_S_4O_6{}^{2-}{}_{(aq)}$$

3. What is the proportion of bleach to water necessary to make water safe to drink?

4. What is the active ingredient in liquid bleach? And the active ingredient in bleach powder?

5. You prepare a 10% v/v diluted bleach solution from a stock bleach solution. Then, you titrate 30mL of diluted bleach solution with 0.025M sodium thiosulfate, reaching the equivalency point after adding 20mL of thiosulfate. Calculate the molarity of the diluted bleach solution and the molarity of the stock bleach solution.

STUDENT INFO	
Name: ____________________	Date: __________

Results
EXPERIMENT

Redox titration of hypochlorite in bleach

		Trial	Exact		
			1	2	3
1	Initial buret reading (mL)	________	________	________	________
2	Final buret reading (mL)	________	________	________	________
3	$Na_2S_2O_3$ added (mL)	________	________	________	________
4	Moles $Na_2S_2O_3$		________	________	________
5	Moles NaOCl		________	________	________
6	[NaOCl] (diluted) (M)		________	________	________
7	[NaOCl] (original) (M)		________	________	________

Average Molarity,[NaOCl] (M)	________
Unknown number (if given)	________

STUDENT INFO

Name: ______________________ Date: __________

Post-lab Questions

Redox titration of hypochlorite in bleach

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1. Use your results to calculate the mass percent assuming a 1.0 g/mL density, comparing your result with the mass percent listed in commercial bleach (5.25% v/v).

2. You prepare a bleach dilution by adding eight drops of concentrated bleach (5.25% v/v) in a water gallon. Calculate the new concentration (% m/m) of the diluted bleach solution.